

3. Equilibrium of a Particle

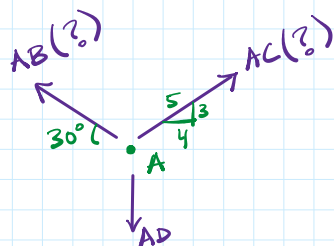
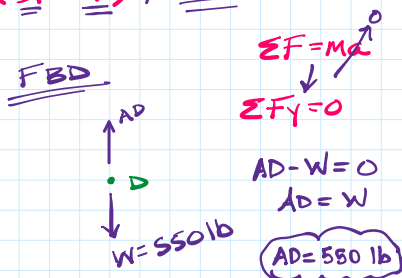
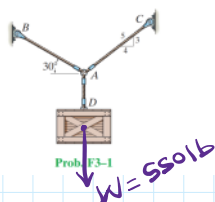
Tuesday, August 20, 2024 8:42 AM

* Analyze particles that are in equilibrium ($\vec{a} = 0$). Constant velocity or @ rest.

* $\vec{F} = m\vec{a} \rightarrow \Sigma \vec{F} = 0$

* Spring $\rightarrow F_s = k(l - l_0)$, FBD

F3-1. The crate has a weight of 550 lb. Determine the force in each supporting cable.



$\Sigma F_y = 0$ $\Sigma F_x = 0$

$AB_x = -AB \cos(30^\circ)$

$AB_y = AB \sin(30^\circ)$

$AC_x = \frac{4}{5} AC$

$AC_y = \frac{3}{5} AC$

$\Sigma F_x = 0$

$AC_x + AB_x = 0 \rightarrow \frac{4}{5} AC - \frac{\sqrt{3}}{2} AB = 0$ (1)

$\Sigma F_y = 0$

$AC_y + AB_y + AD = 0 \rightarrow \frac{3}{5} AC + \frac{1}{2} AB - 550 = 0$ (2)

$\frac{3}{5} AC + \frac{1}{2} AB = 550$ (2a)

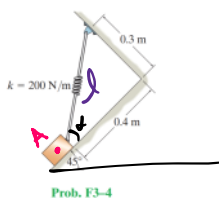
$$\begin{bmatrix} \frac{4}{5} & -\frac{\sqrt{3}}{2} \\ \frac{3}{5} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} AC \\ AB \end{bmatrix} = \begin{bmatrix} 0 \\ 550 \end{bmatrix}$$

A · X = b

$AB = 478.46 \text{ lb}$

$AC = 516.74 \text{ lb}$

F3-4. The block has a mass of 5 kg and rests on the smooth plane. Determine the unstretched length of the spring.



l_0 - unstretched length.

$l = 0.5 \text{ m}$

$F_{\text{spring}} = k(l - l_0)$, $W = mg$

Geometry makes problem more challenging

* Mass is in equilibrium ($\Sigma F = 0$)

* Decompose vectors into x & y components

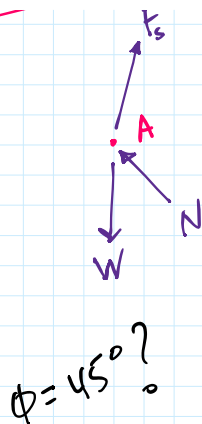
$\Sigma F_x = 0$ $\Sigma F_y = 0$



$F_s = F \cdot \cos \theta$

$\theta = 45^\circ + \alpha$

$\tan \alpha = \frac{1}{\sqrt{3}} = 30.96^\circ$



$$\sum F_x = 0 \quad \sum F_y = 0$$

$$F_{sx} = F_s \cdot \cos \theta$$

$$F_{sy} = F_s \cdot \sin \theta$$

$$\theta = 45^\circ + \alpha$$

$$\tan \alpha = \left(\frac{0.3}{0.4} \right) = 36.86^\circ$$

$$\theta = 45^\circ + 36.86^\circ$$

$$\theta = 81.86^\circ$$

Solve using calculator

$$A \cdot x = b$$

$$\begin{bmatrix} \cos(81.86^\circ) & -\sin(45^\circ) \\ \sin(81.86^\circ) & \cos(45^\circ) \end{bmatrix} \begin{bmatrix} F_s \\ N \end{bmatrix} = \begin{bmatrix} 0 \\ 49.1 \end{bmatrix}$$

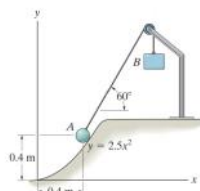
$$\sum F_x = F_{sx} - N_x = 0 \rightarrow F_s \cos(81.86^\circ) - N \sin(45^\circ) = 0 \quad (1)$$

$$\sum F_y = F_{sy} + N_y - W = 0 \rightarrow F_s \sin(81.86^\circ) + N \cos(45^\circ) = 49.1 \quad (2)$$

$$F_s = 43.3 \text{ N} \quad N = 8.6 \text{ N}$$

$$F_s = K(l - l_0) \rightarrow 43.3 = 200(0.5 - l_0) \quad l_0 = 0.28 \text{ m}$$

3-37. A 4-kg sphere rests on the smooth parabolic surface. Determine the normal force it exerts on the surface and the mass m_B of block B needed to hold it in the equilibrium position shown.

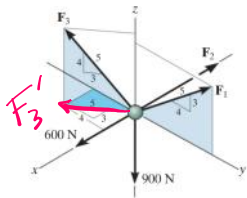


Prob. 3-37

$$m = 4 \text{ kg} \quad W = mg$$

$$m_B = ? \quad N_A = ?$$

F3-7. Determine the magnitude of forces F_1 , F_2 , F_3 , so that the particle is held in equilibrium.



Prob. F3-7

$$F'_3 = \frac{3}{5} F_3$$

Known Forces
 $600 \hat{i} + 0 \hat{j} + 0 \hat{k} \text{ N}$
 $0 \hat{i} + 0 \hat{j} - 900 \hat{k} \text{ N}$

$$\vec{F}_1 = 0 \hat{i} + \frac{4}{5} F_1 \hat{j} + \frac{3}{5} F_1 \hat{k}$$

$$\vec{F}_2 = -F_2 \hat{i} + 0 \hat{j} + 0 \hat{k}$$

$$\vec{F}_3 = \frac{3}{5} \left(\frac{3}{5} F_3 \right) \hat{i} - \frac{4}{5} \left(\frac{3}{5} F_3 \right) \hat{j} + \frac{4}{5} F_3 \hat{k}$$

$$\sum F_x = 0$$

$$600 - F_2 + \frac{9}{25} F_3 = 0$$

$$\sum F_y = 0$$

$$\frac{4}{5} F_1 - \frac{12}{25} F_3 = 0 \rightarrow F_1 = \frac{5}{4} \left(\frac{12}{25} \right) F_3 = \frac{3}{5} F_3$$

$$\sum F_z = 0$$

$$-900 + \frac{3}{5} F_1 + \frac{4}{5} F_3 = 0$$

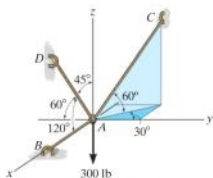
$$900 = \frac{3}{5} \left(\frac{3}{5} F_3 \right) + \frac{4}{5} F_3 \rightarrow 900 = \left(\frac{9}{25} + \frac{4}{5} \right) F_3$$

$$900 = \frac{29}{25} F_3 \rightarrow F_3 = 900 \left(\frac{25}{29} \right) \rightarrow F_3 = 775.86 \text{ N}$$

$$F_1 = \frac{3}{5} (775.86) \rightarrow F_1 = 465.52 \text{ N}$$

$$F_2 = 600 + \frac{9}{25} (775.86) \rightarrow F_2 = 879.30 \text{ N}$$

F3-10. Determine the tension developed in cables AB, AC, and AD.



Prob. F3-10

$$AD_x = AD \cos(120^\circ) = -0.5 AD$$

$$AD_y = -AD \cos(60^\circ) = -0.5 AD$$

$$AD_z = AD \cos(45^\circ) = 0.707 AD$$

$$AC_x = AC \cos(60^\circ) \sin(30^\circ) = 0.25 AC$$

$$AC_y = AC \cos(60^\circ) \cos(30^\circ) = 0.433 AC$$

$$AC_z = AC \sin(60^\circ) = 0.866 AC$$

$$AB_x = AB$$

$$AB_y = 0$$

$$AB_z = 0$$

$$\sum F_x = 0$$

$$\textcircled{1} -0.5 AD - 0.25 AC + AB = 0$$

$$\sum F_y = 0$$

$$\textcircled{2} -0.5 AD + 0.433 AC + 0 = 0 \rightarrow$$

$$AD = 0.866 AC$$

...substitute

$$\sum F_y = 0$$

$$(2) -0.5AD + 0.433AC + 0 = 0 \rightarrow$$

$$AD = 0.866AC$$

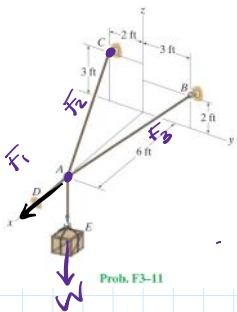
$$\sum F_z = 0$$

$$(3) 0.707AD + 0.866AC + 0 = 300$$

Substitute

$$AB = 139 \text{ lb} \quad AC = 203 \text{ lb} \quad AD = 176 \text{ lb}$$

F3-11. The 150-lb crate is supported by cables AB, AC, and AD. Determine the tension in these wires.



Prob. F3-11

Unit vectors

$$\underline{AC} = \frac{-6\hat{i} - 2\hat{j} + 3\hat{k}}{\sqrt{(-6)^2 + (-2)^2 + 3^2}}$$

$$AC = -\frac{6}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{3}{7}\hat{k}$$

$$\vec{F}_2 = F_2 \cdot \left\langle -\frac{6}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{3}{7}\hat{k} \right\rangle$$

Unit Vector

$$\underline{AB} = \frac{-6\hat{i} + 3\hat{j} + 2\hat{k}}{\sqrt{(-6)^2 + (3)^2 + (2)^2}}$$

$$AB = -\frac{6}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{2}{7}\hat{k}$$

$$\vec{F}_3 = F_3 \cdot \left\langle -\frac{6}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{2}{7}\hat{k} \right\rangle$$

$$\sum F_x = 0$$

$$F_1 - \frac{6}{7}F_2 - \frac{6}{7}F_3 = 0$$

$$\sum F_y = 0$$

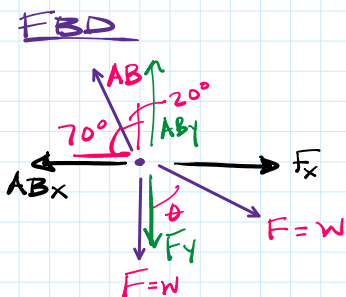
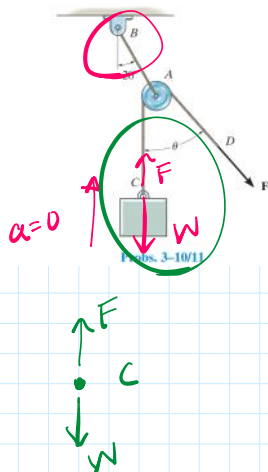
$$-\frac{2}{7}F_2 + \frac{3}{7}F_3 = 0$$

$$\sum F_z = 0$$

$$\frac{3}{7}F_2 + \frac{2}{7}F_3 - W = 0$$

3-10. The block has a weight of 20 lb and is being hoisted at uniform velocity. Determine the angle θ for equilibrium and the force in cord AB.

3-11. Determine the maximum weight W of the block that can be suspended in the position shown if cords AB and CAD can each support a maximum tension of 80 lb. Also, what is the angle θ for equilibrium?



$$\sum F_x = 0$$

$$AB \cos(70^\circ) = F \sin \theta$$

$$AB \cos(70^\circ) = W \sin \theta \quad (1)$$

$$\sum F_y = 0$$

$$AB \sin(70^\circ) = F + F \cos \theta$$

$$(2) \quad AB \sin(70^\circ) = W(1 + \cos \theta)$$

$$AB = \frac{W \sin \theta}{\cos(70^\circ)}$$

$$\cancel{W \sin \theta} \cdot \tan(70^\circ) = \cancel{W} (1 + \cos \theta)$$

$$\sin \theta \cdot \tan(70^\circ) = 1 + \cos \theta$$

$$\theta = 0.7 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right)$$

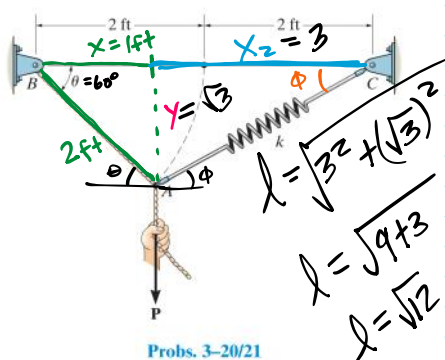
$$\theta = 40.1^\circ$$

Solved graphically

$$AB = \frac{20 \cdot \sin(40.1^\circ)}{\cos(70^\circ)}$$

$$AB = 37.67 \text{ lb}$$

3-21. Determine the unstretched length of spring AC if a force $P = 80$ lb causes the angle $\theta = 60^\circ$ for equilibrium. Cord AB is 2 ft long. Take $k = 50$ lb/ft.



Probs. 3-20/21

$$x = 2 \cos 60^\circ$$

$$x_2 = 3 \text{ ft}$$

$$x = 1 \text{ ft}$$

$$\tan \phi = \frac{y}{x_2}$$

$$x^2 + y^2 = 2^2$$

$$y = \sqrt{3} = 1.73 \text{ ft}$$

$$\phi = \tan^{-1}\left(\frac{1.73}{3}\right)$$

$$\phi = 29.97^\circ$$

$$\sum F_x = 0$$

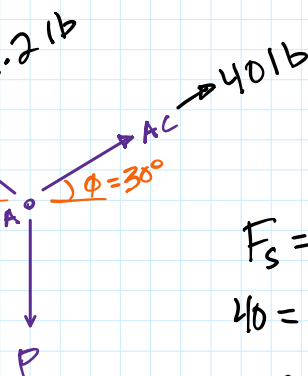
$$AC_x - AB_x = 0$$

$$AC_x = AB_x$$

(2)

$$AC \cdot \cos(30^\circ) - AB \cos(60^\circ) = 0$$

$$AC \cdot \left(\frac{\sqrt{3}}{2}\right) - AB \left(\frac{1}{2}\right) = 0$$



$$F_s = k(l - l_0)$$

$$40 = 50(\sqrt{12} - l_0)$$

$$0.8 = \sqrt{12} - l_0$$

$$l_0 = \sqrt{12} - 0.8$$

$$l_0 = 2.6 \text{ ft}$$

$$\sum F_y = 0$$

$$AB_y + AC_y - P = 0$$

$$AB_y + AC_y = P$$

(1)

$$AB \sin(60^\circ) + AC \sin(30^\circ) = P$$

$$AB \left(\frac{\sqrt{3}}{2}\right) + AC \cdot \left(\frac{1}{2}\right) = 80$$

AB	AC	RHS
$\begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$	$\begin{bmatrix} 80 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix} \begin{bmatrix} AB \\ AC \end{bmatrix} = \begin{bmatrix} 80 \\ 0 \end{bmatrix}$$

$$\text{rref} \left(\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} & 80 \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \end{bmatrix} \right)$$

calculator
→

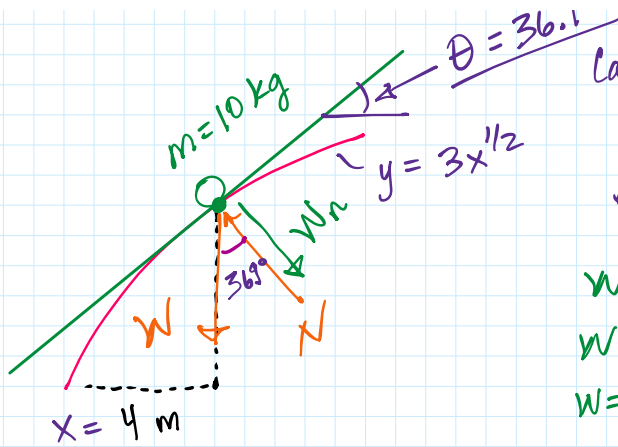
$$\begin{bmatrix} 1 & 0 & \frac{AB}{AC} \\ 0 & 1 & \frac{AC}{AB} \end{bmatrix}$$

1.73

1.2

$$\theta = 36.9^\circ$$

calculate
h.o. normal

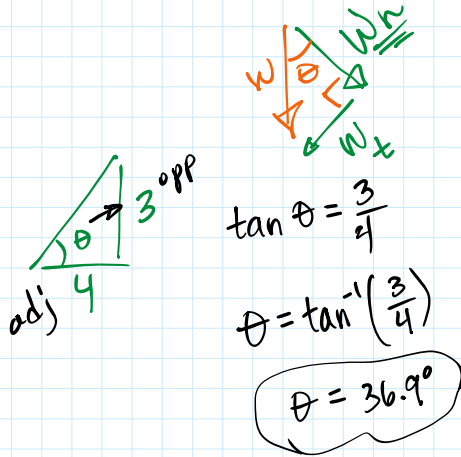
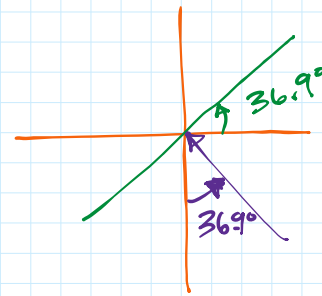


Calculate the normal force that the surface exerts on the particle.

$$W = mg$$

$$W = (10 \text{ kg})(9.81 \text{ m/s}^2)$$

$$W = 98.1 \text{ N}$$



$$\frac{d}{dx} (3x^{1/2}) = \frac{3}{2} x^{-1/2}$$

slope @ $x = 4 \text{ m}$

$$\frac{3}{2} (4)^{-1/2} = \frac{3}{4}$$

rise over run

$$N = W_n = W \cos(36.9^\circ) = 98.1 \cos(36.9^\circ)$$

$$N = 78.4 \text{ N}$$

Decompose

F_x & F_y

or

F_t & F_n

$$F_t = F \cos(32^\circ)$$

$$F_n = F \sin(32^\circ)$$

$$F_x = F \cos(60^\circ)$$

$$F_y = F \sin(60^\circ)$$

